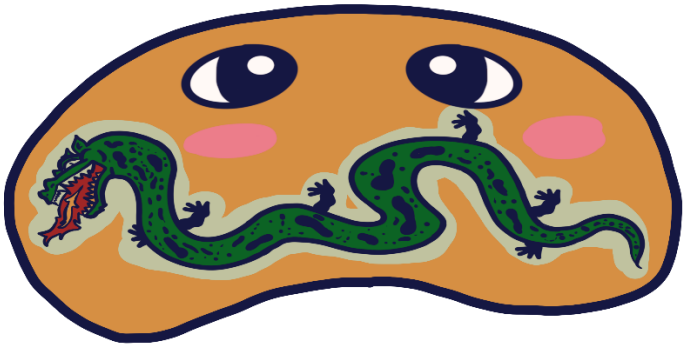
# Project Name

**Sponsored by: Dr. Powell and Dr. Shumaker**
mpowell@uab.edu, klshumak@uab.edu
**Team**
ATP and Me

**2/2/2025**



## Executive Summary

The project is a game named "Electronic Transport Game" sponsored by the UAB Biology Department professors, Dr.Powell and Dr. Shumaker, and created by the "ATP and Me" group is an interactive way that is easy and educational for early biology students to learn and study a topic that is considered hard to grasp. The goal of the project is that students should be able to fill in a blank diagram of the Oxidative Phosphorylation, and more specifically the Electron Transport process. Interactivity of the entire process will be one of the strongest focuses of this project. Students tend to be overwhelmed when learning this portion of Biology as they are introduced to many new concepts all at once, therefore providing students with the ability to engage with the Electron Transport process will help them understand the concepts more effectively.

The project will utilize GitHub and Godot to build the project. The minimum deliverable will consist of a guided game mode, which is the learning process, and if time permits an unguided game mode will be included in deliverables as well, which is an overall review of the content. The guided game mode will teach students about the Electron Transport process and its various steps, as well as teach them how to play the game. The focus of the project will be the guided mode, as Dr. Powell informed us that many students focus on that mode on other Biology games that the department has used. However, as stated previously the team would like to add in an unguided mode to give students the extra option in their interactive studies. The group "ATP and Me" will attempt to meet with the sponsors biweekly or as requested, and the group would learn and understand the special topic of oxidative phosphorylation. This is to ensure that the team is following the wishes of the sponsors as closely as possible so that "Electron Transport Game" will be successful. The aim of this project is to help students and meeting with the sponsors to ensure that our information is the most accurate it can't be and that the game is going to be helpful for students and it is of utmost importance for the group.

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## Statement of the Problem

The problem is that students are introduced to a lot of new concepts very quickly when they begin learning about the Electron Transport Process of oxidative phosphorylation, which causes students to struggle. We are trying to solve it because we want to provide students with an engaging and fun experience while also helping them learn. Many things are being done to help students learn the Electron Transport Process, such as quizzes and even a model building simulator.

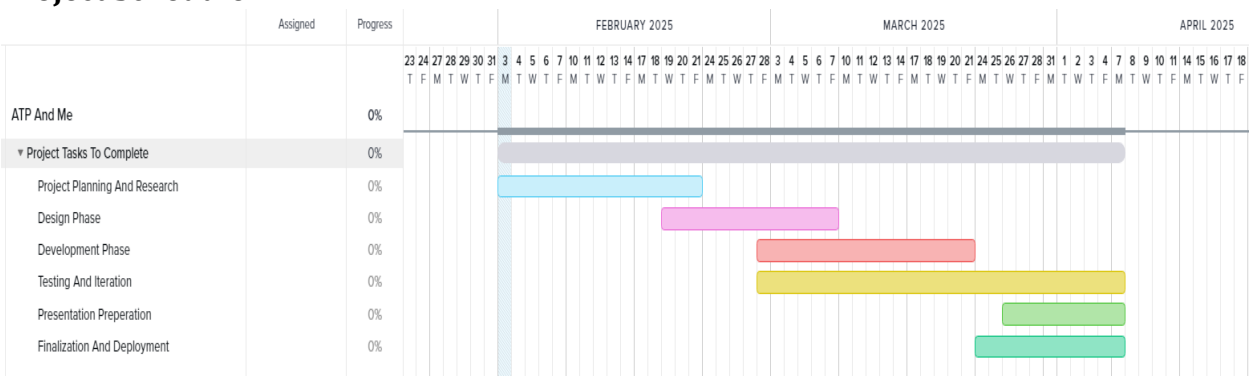
## Objectives

Our proposed project is a game, this fills a role in the domain that we chose because, while there are sources to help students learn the Electron Transport Process, none are as engaging as a game designed to retain students' attention while they are learning. This project hopes to take advantage of the entertainment factor of gaming to increase students' learning on this subject like nothing else does.

## Technical Approach

The engine we are going to use for this project is going to be Godot. The Godot API uses a mix of C++, C#, and Godot's own GDScript.

## Project Schedule




## Deliverables

Our approach is to at least create a game with a guided game mode. The guided game mode will serve to teach students the material as they play, as well as teach students how to play the game. If time permits, we would like to create an unguided mode as well, so that students could challenge themselves extra if they wish.

**Budget**
No budget.

**Communication and Coordination with Sponsor**

We will meet with our sponsors biweekly or as requested to ensure that our progress is going in the right direction, as well as to plan future steps of the project.

**Special Topics**

Our special topic is also the focus of the game, the Electron Transport Process of Oxidative Phosphorylation.